

Homework — 1.3.1 Compression, Encryption and Hashing

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.3.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. A backup program handles the files listed below.

(a) Complete the table to show whether each file must use **lossy** or **lossless** compression. [2] (AO2)

File	Lossy or lossless?
Software installer kept as a backup	
Photograph displayed on a web page	
Folder of source-code text files	
Music streamed to a phone	

(b) Give **one** reason why your choice for the source-code text files is essential. [1] (AO2)

2. A student applied **run-length encoding (RLE)** to this row of pixels:

R R R R G G G R R B B B B

The student wrote the encoding 3R 3G 2R 5B, which contains **two** errors. Identify both errors and write the fully correct encoding. [2] (AO2)

3. A different row of pixels is R G R G R G R G. Explain why applying RLE to this row is a poor choice, referring to the size of the encoded output. [2] (AO2)

4. Complete the paragraph about **dictionary coding** using words from the word bank. Not all the words are used. **[2] (AO1)**

In dictionary coding, frequently occurring words or patterns in the text are stored once in a (a) _____. Each occurrence in the document is replaced by a short (b) _____ that points to the matching entry. Because each reference takes less space than the pattern it replaces, the compressed file is (c) _____ than the original. The dictionary must be stored or transmitted with the compressed data so the references can be looked up during (d) _____.

Word bank: decompression · dictionary · index · key · larger · pixel · smaller

5. Alice wants to send a confidential file to Bob, whom she has never met, so that only Bob can read it. Number the steps of the **asymmetric encryption** process from 1 to 6 in the correct order. **[3] (AO2)**

Step	Order (1–6)
Alice encrypts the file using Bob's public key.	
Bob generates a matching public and private key pair.	
Bob uses his private key to decrypt the file.	
Alice obtains a copy of Bob's public key.	
The encrypted file is sent to Bob across the internet.	
Bob makes his public key openly available.	

6. A website stores a hash of each user's password rather than the password itself.

(a) State **one** property of a hash function that keeps the passwords safe even if the database of hashes is stolen. **[1] (AO1)**

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(b) Describe how the system checks that a user has entered the correct password at login, without ever storing the plaintext password. **[1] (AO2)**

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Part B — Retrieval: keep it fresh (~6 marks)

7. (1.2.2) Explain **one** difference between a **compiler** and an **interpreter**. **[2] (AO1)**

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8. (1.1.1) State what is held in the **MAR** and the **MDR** during the fetch stage of the fetch–decode–execute cycle. [2] (AO1)

9. (1.2.1) State the purpose of the **scheduler** in an operating system, and give **one** aim it tries to achieve. [2] (AO1)
